Building

The Buildings are the structures that are built with BuildingElements. It is used for collecting and counting the BuildingElements and it is used for some algorithms' calculation, validation process caching etc.

Because of that every building algorithms' has different needs every building algorithms has its own unique type of buildings:

- Advanced Grid Building
- Basic Grid Building
- Free Building
- SnapPoint Based Building (Experimental!)

All of them are inherited from the base Building class which is inherited from the MonoBehavior.

The Object structure in your hierarchy should follow the following rules to ensure that the Advanced Building System can work properly. (Use the SceneHelper tool for validate the Hierarchy)

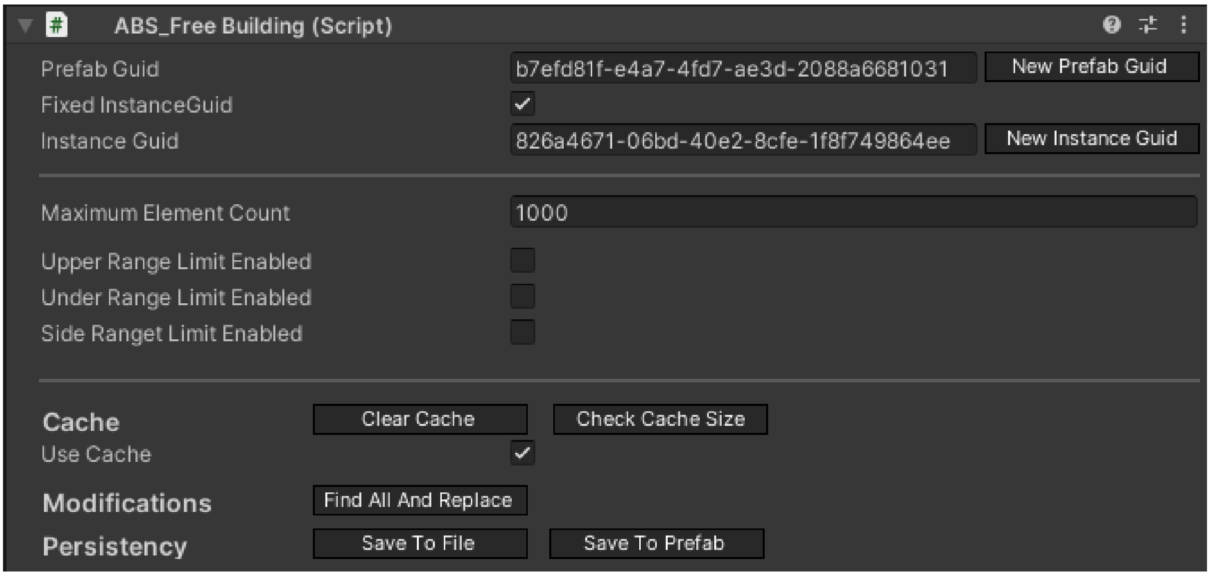
1. Every BuildingElement is assigned with a building algorithm settings and every Building should have only BuildingElements with the matching type of algorithm.
2. Every Building should be under a GameObject with BuildingParent script in the Hierarchy and every object under a BuildingParent should be only GameObjects with Building script.
3. Every BuildingElement should be under a GameObject with Building script in the Hierarchy and every object under a Building should be a GameObject with BuildingElement script.

In the case of Basic Grid Building and Free Building the algorithms use a global building so the same Building will be used every time anywhere in the scene. For that the BuildingParent object has dedicated Buildings for that case.

In the case of Advanced Grid Building or SnapPoint Based the algorithm will create a new building if a new BuildingElement will be placed without snapping to another element. If an element is snapping to another one the new element will be placed under the snapping target's parent building.

If every element is removed from a Building so it has left zero child BuildingElement the Building will be destroyed. This is not true for global buildings.

Property of the building scripts



At this moment it is not recommended to build a building by hand. Create it in a test scene and use the Save To Prefab button on the Buildings' inspector view. This feature is available on every Building type. But if you create your Building ensure that the BuildingElements has a valid position. Also in the case of Advanced Grid Building ensure that you refresh the Modifier after you placed the first Element under the Building.

Maximum Element Count:

The BuildingElement's count can not be higher than the set number. If you try to build an element to a building with a maxed out element count the placement of the new element will be blocked. In case of DragBuilding the amount of elements above the maximum count will be blocked.

For example in case of 100 maximum element count and the Building has 90 elements then only 10 elements can be built so in case of drag building 10 elements will be allowed and every element above that 10 will be blocked.

Range Limits

3 type of limit are currently available:

- Upper Range Limit
- Under Range Limit
- Side Range Limit

The limits are for validation purposes. If any of them are enabled the position of the element will be checked during the position validation process.

If the Upper Limit is enabled the element's position can not be above the set limit.
If the Under Limit is enabled the element's position can not be under the set limit.
If the Side Limit is enabled the element's position can not be outside the set value's distance. This is checked only horizontally.

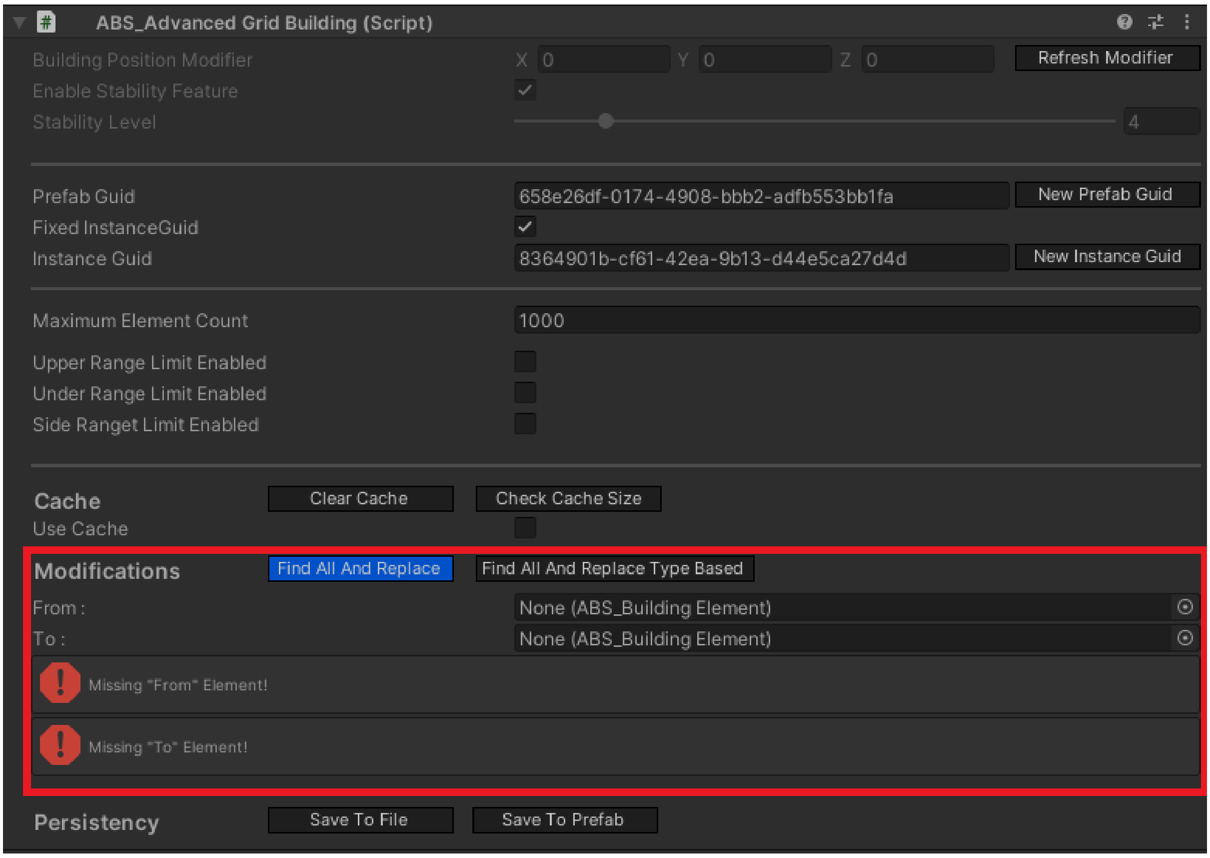
The Under Limit should be negative or zero.
The Side and the Upper limits value should be zero or above.
Note that every position is calculated from the Building's local zero (0,0,0) position.

Cache

The Buildings are caching some data to increase the performance of the positioning algorithms. The cache increases the memory usage of the Buildings so the cache can be cleared, enabled or disabled any time. Note that disabling the case implicates the clearing of the cache's data.

Find All And Replace

The Find and Replace button is an editor feature which will call the FindAllAndReplaceElements function on the building which is available during runtime if you need that functionality. The FindAllAndReplaceElements feature needs two BuildingElement. The "From" element will be searched in the children of the Building which will be replaced by the "To" element.



That it will replace every BuildingElements that match (the Guids are equal) with the provided "From" element.

The FindAllAndReplaceElements feature has some requirement :

- The provided BuildingElements must be not null.
- The elements can not be the same element so the guids have to be different.

- The elements’ main algorithm should be matching with the Building’s type
- The elements’ AdvancedGridType and AdvancedGridAxisType should be matching in case of AdvancedGridBuilding

If the requirements are not fulfilled the elements can’t be replaced.

Find All And Replace Type Based

This feature is only available for the AdvancedGridBuildings.
This feature is similar to the Find All And Replace. The mentioned other feature is based on prefabs but this one works with types.

It works in that way an element should be provided on the UI. For the Execute all of the elements that have the same AdvancedGridType as the given element will be replaced to that element.

Persistency

You can save the Building into a file with the Save To File. It will save a json file with the save data of the Building that is used during the save/load processes.

You can use the BuildingLoader tool for loading a building from json data or the buildings can be loaded from scripts too. For more information check the persistence documentation.

IBuildingExternalInterface

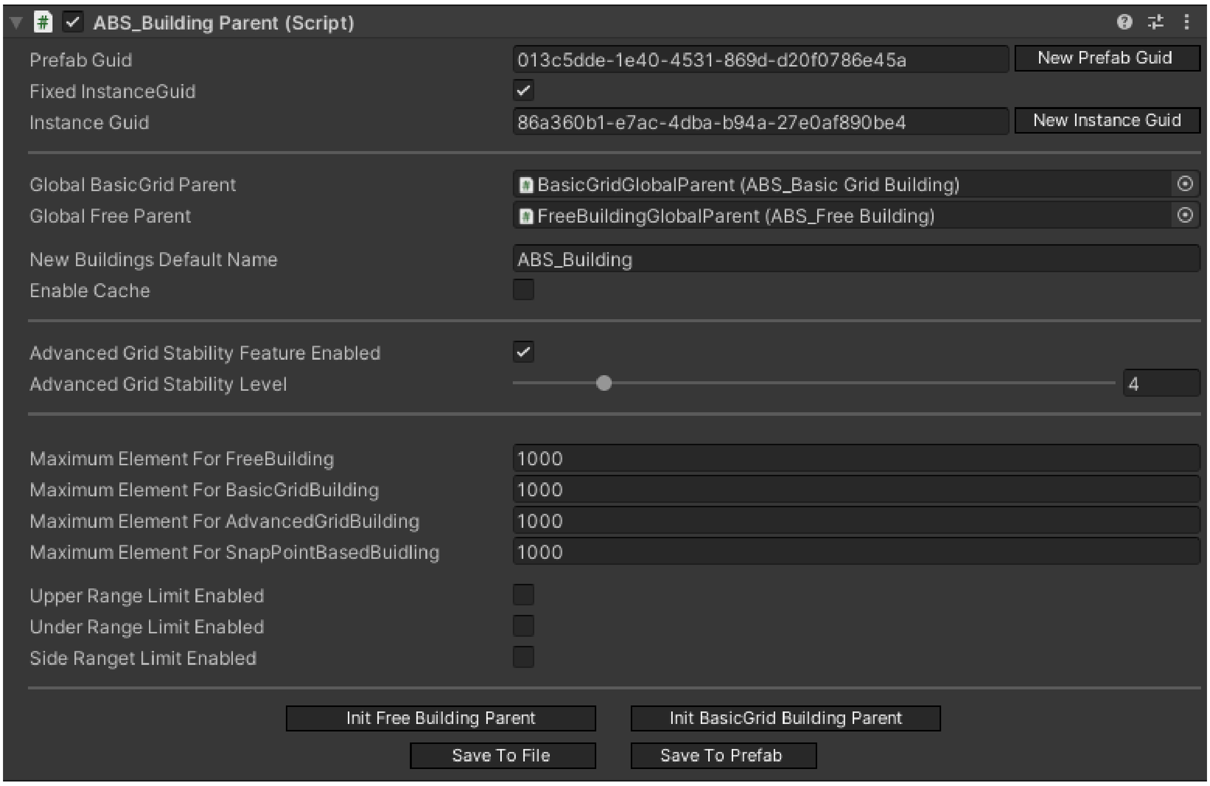
The IBuildingExternalInterface is a collection of functions that is available to the developers who's using this asset to deal with the Buildings. Every other public function is for the Advanced Building System. The goal of this interface is to help the developers and make it easier to know what options they have. Any other function call on the Buildings that is not included by this interface is not recommended.

Note that the Buildings are inherited from the SaveableMonobehaviour. Its function can be used on the Buildings to reach the persistence functionalities.

- **Getters/Setters**
    - LayerCollection
    - Elements
    - PositionSearchAlgorithmType
- **Add**
    - AddBuildingElement (2 overload)
        - With Position
        - With Position + Rotation
- **Remove**
    - RemoveBuildingElement
- **Search / Checks / Find**
    - FindBuildingElement
    - ContainsBuildingElement
    - GetPositionOfBuildingElement
    - FindAllBuildingElement (3 overload)
        - Based on BuildingElement’s Guid
        - Based on BuildingElement’s AreaType
    - FindAllPreBuiltBuildingElement
    - FindAllFoundationBuildingElement
        - Inverse option = Find all not foundation
- **Element Modifications**
    - FindAllAndReplaceElements
    - MakePreBuilt
    - RemovePreBuilt
- **Material Modifications**
    - SetMaterialToDefault
    - SetMaterialBasedOnState
- **Cache**
    - ClearCache
    - EnableCache
    - DisableCache

Building Parent

Every Building should be under a GameObject with a BuildingParent script. This script will manage and create the Buildings.



Properties of the Building Parent

Prefab Guid is used for identifying the object from the same prefabs.

FixedInstanceGuid If this is true then the Instance Guid can be set. The set InstanceGuid will be used by the algorithm and it will not be changed. If this boolean is false then during the construction of the object a new InstanceGuid will be generated automatically.

InstanceGuid is used for identifying the instances of the objects.

Global BasicGrid Parent is the global Basic Grid Building parent that is used for every new BuildingElement built with the BasicGrid Building algorithm.

Global Free Parent is the global Free Building parent that is used for every new BuildingElement built with the Free Building algorithm.

New Buildings Default Name will be the name of the newly created Buildings.

Enable Cache is a boolean variable and if it is true then the newly created Building's cache will be enabled. The cache will speed the algorithms up but increase the memory usage of the algorithms.

Advanced Grid Stability Feature Enabled is boolean and if it is true then the stability feature will be enabled for the newly created AdvancedGridBuildings.

AdvancedGrid Stability Level is the maximum stability level that will get the stable elements for the AdvancedGridBuilding if the Stability feature is enabled.

Maximum Element for … variables will be the maximum element count for the newly created Buildings for the specific type of Building.

Upper Range Limit Enabled

When this boolean is true then the upper limit range will be enabled on the Buildings. Which means that the building of an element will be blocked if the position of the element is above the set value.

Under Range Limit Enabled

When this boolean is true then the under limit range will be enabled on the Buildings. Which means that the building of an element will be blocked if the position of the element is under the set value. This value should be always negative.

Side Range Limit Enabled

When this boolean is true then the side limit range will be enabled on the Buildings. Which means that the building of an element will be blocked if the position of the element is outside of the set value. In this case the distance is calculated with 0 y values. So the distance will be calculated between the element positions (x, 0, z) and the local zero (0,0,0) to only calculate the horizontal distance.

Note that the range limits are calculated from the local zero (0,0,0) position from the Building.

Other options

Init Free Building Parent

If the Global Free Parent is null then the button press will create a Free Building and set it to that property.

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Persistency

Just like the Buildings the Building Parent can be saved too into a file in a JSON format. Or it can be saved into a prefab as well.
In case of the save of the BuildingParent all of its Buildings data with their BuildingElements will be saved. So the whole Building parent can be saved and loaded at one time.
For the load during the development process the Building Loader developer tool can be used as explained in the tools documentation.
During the game's runtime the BuildingParent provides a set of functions for saving and loading automatically. For more information check the Persistency documentation.